

# INTRODUCTION TO MICROSOFT POWER APPS

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# Agenda

- What is Power Platform?
- Fusion development
- Power Apps introduction
  - What is Power Apps?
  - Why use Power Apps?
  - What can you build
  - Power Apps components
    - Canvas apps
    - Dataverse
    - Model-driven apps
    - Integration with Power Automate
- Hackathon labs: Power App in a day

# What is Power Platform?

- Collection of low/no-code tools for rapidly building custom business apps
  - Turns business needs into secure, connected applications
- Main components
  - Power Apps
    - Builder for creating custom mobile and web business applications
  - Power Automate
    - Workflow service that automates repetitive tasks and orchestrates activities across services using connectors
  - Power BI
    - Provides interactive data visualization Business Intelligence tools
  - Power Pages
    - Platform for designing, hosting and managing secure business websites
  - Copilot Studio
    - Tool for building AI-powered conversational agents

# Fusion development

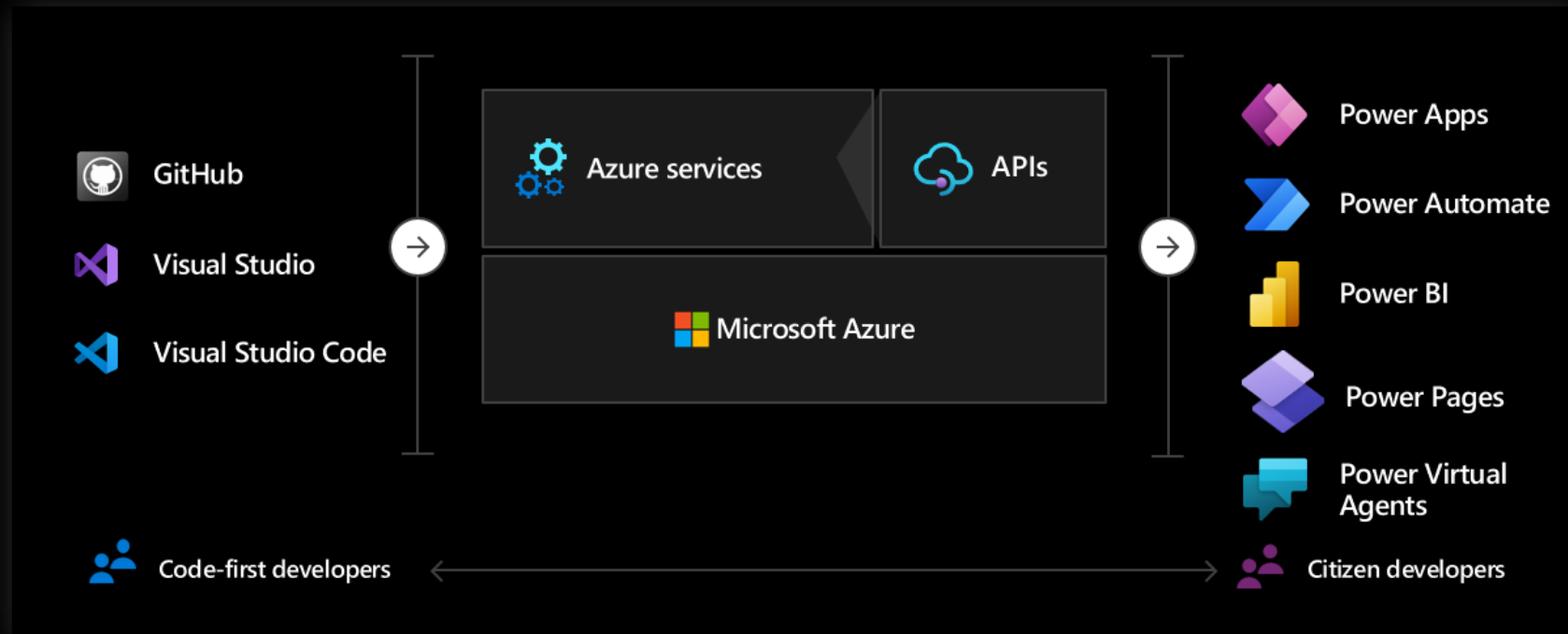
- “Distributed and multidisciplinary digital business teams that blend technology and other types of domain expertise.”
  - 2019 Gartner Digital Business Teams Survey
- Enables building apps
  - Better
  - Faster
- Goal: Scale application development through low-code
  - Empower people with different skill sets

# Fusion team roles

- Citizen (low-code) developers
  - Non-technical
  - Knowledgeable about ways to improve business processes
  - Use visual drag/drop tools
- Professional developers
  - Highly technical
  - Write code for custom connectors, integrating w/external data, applying business logic
  - Use tools like Visual Studio/VS Code
- IT professionals/DevOps engineers
  - Responsible for application lifecycle management (ALM) of business apps

# Fusion team development

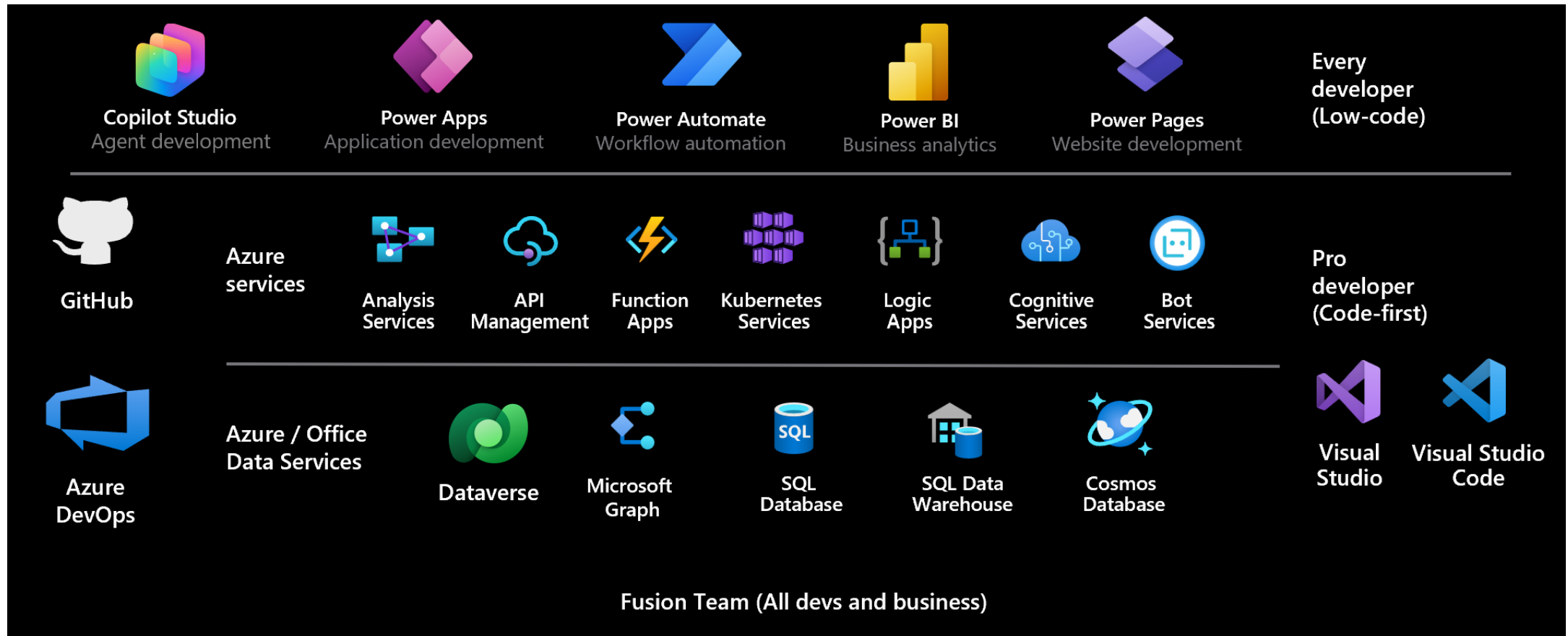
## Fusion Team Development



# Citizen developer tools

- Designed to make the citizen developer more productive
- Created
  - By professional developer/IT professional
  - Within a secure and governed environment
- Should be usable by everyone for low-code tasks
  - Not limited to citizen developers
  - E.g., Visual tooling for creating apps

# Fusion team development tools





# What is Power Apps?

- Low-code platform for providing the business app user experience
  - Mobile & web
- Software suite consisting of
  - Apps
  - Services
  - Data platform
  - Connectors

# Why Power Apps?

- Build quickly
  - Use visual designers, formulas, and reusable components
- Connect data using
  - Dataverse
  - Many cloud or on-prem sources
- Run anywhere
  - Web browser
  - Mobile devices
    - Phone (iOS and Android)
    - Tablets

# What can you build?

Look for manual, repetitive, data-driven processes that need a better experience.

## Replace manual work

Forms, spreadsheets, paper

## Track activities

Assets, requests, issues

## Modernize legacy apps

Departmental tools

## Capture field data

Inspections, visits, inventory

## Standardize approvals

Rules, routing, status

## Extend existing systems

Focused task experiences

**Candidate test: Is the process manual, repetitive, data-driven, or fragmented across tools?**



# Power Apps components

- Canvas apps
  - Start from a blank canvas, build interface & connect to data sources
- Microsoft Dataverse
  - Data platform for Power Apps that stores and models business data
- Model-driven apps
  - Start with data model & build up to model forms, views & other components
- Plans
  - Copilot-first tool for building end-to-end business solution
- Copilot
  - Create an app & data model using Copilot

# Canvas apps

- Start from blank design surface for building user experience
  - Drag and drop components to build UI
- Connect
  - > 200 data sources
  - Works on all major browsers and mobile devices
- Share
  - Securely within organization
- Embed
  - Directly in SharePoint, Power BI and Teams

# Dataverse

- Store business data
  - Secure cloud-based storage for Power Platform applications
  - Standard and custom tables for business information
- Model relationships
  - Connect tables through relationships
  - Create a shared source of truth across applications
- Manage business logic
  - Business rules help enforce data consistency
  - Calculated and derived values reduce application complexity
- Secure
  - Role-based security controls access to data
  - Supports organizational security boundaries
- Integrate
  - Used by Power Apps, Power Automate, and Dynamics 365
  - Accessible through connectors and APIs
- Scale
  - Designed for enterprise applications and large data volumes
  - Centralized platform for data, automation, reporting, and AI solutions

# Why use Dataverse?

- Zero-Code Security
  - Deeply granular role-based security model
- Instant API Generation
  - Automatically generates secure standardize OData REST API
- Native Power Platform integration
  - Acts as “brain” for Microsoft 365
- Logic and validation
  - Can embed business rules, calculated fields, plug-in logic directly into data layer
- Rich data types and file storage
  - Handles complex data (e.g., images, multi-GB attachments) out of the box

# Checklist for when to use Dataverse

- Only store data that
  - Drives user interaction
  - Triggers automation
  - Requires strict app-level security
- Store in Dataverse if answer is “yes” to any of the following questions:
  - Does a user need to edit the data inside a Power App?
  - Does a Power Automate flow need to trigger the second the data changes?
  - Is this record uniquely tied to security roles managed in Power Platform



# Model-driven apps

- Start from the data model
  - Build on Dataverse tables, relationships, forms, and views
- Generate the application
  - Navigation, forms, and views are generated from app components
  - Focus on data and business processes rather than pixel-perfect layout
- Manage processes
  - Business process flows guide users through work
  - Business rules enforce data consistency
- Secure
  - Role-based security controls access to data and application features
- Share
  - Securely within the organization
- Extend
  - Add charts, dashboards, Power Automate flows, and custom pages

# Power Automate

- Automate processes
  - Trigger workflows from events, schedules, or user actions
  - Reduce manual and repetitive tasks
- Connect systems
  - Integrate Microsoft 365, Power Platform, and external services
  - Hundreds of prebuilt connectors available
- Manage approvals
  - Route requests to approvers automatically
  - Track decisions and outcomes
- Orchestrate workflows
  - Combine data, business rules, and actions
  - Coordinate activities across multiple systems
- Respond to events
  - Start flows when records are created or updated
  - Send notifications and update related systems
- Extend solutions
  - Add automation to Power Apps and Dataverse
  - Support end-to-end business processes

# Canvas vs. Model-Driven apps

## Which app type should you use?

Choose based on the experience the process needs, not which designer looks easier.

**START WITH** The user experience

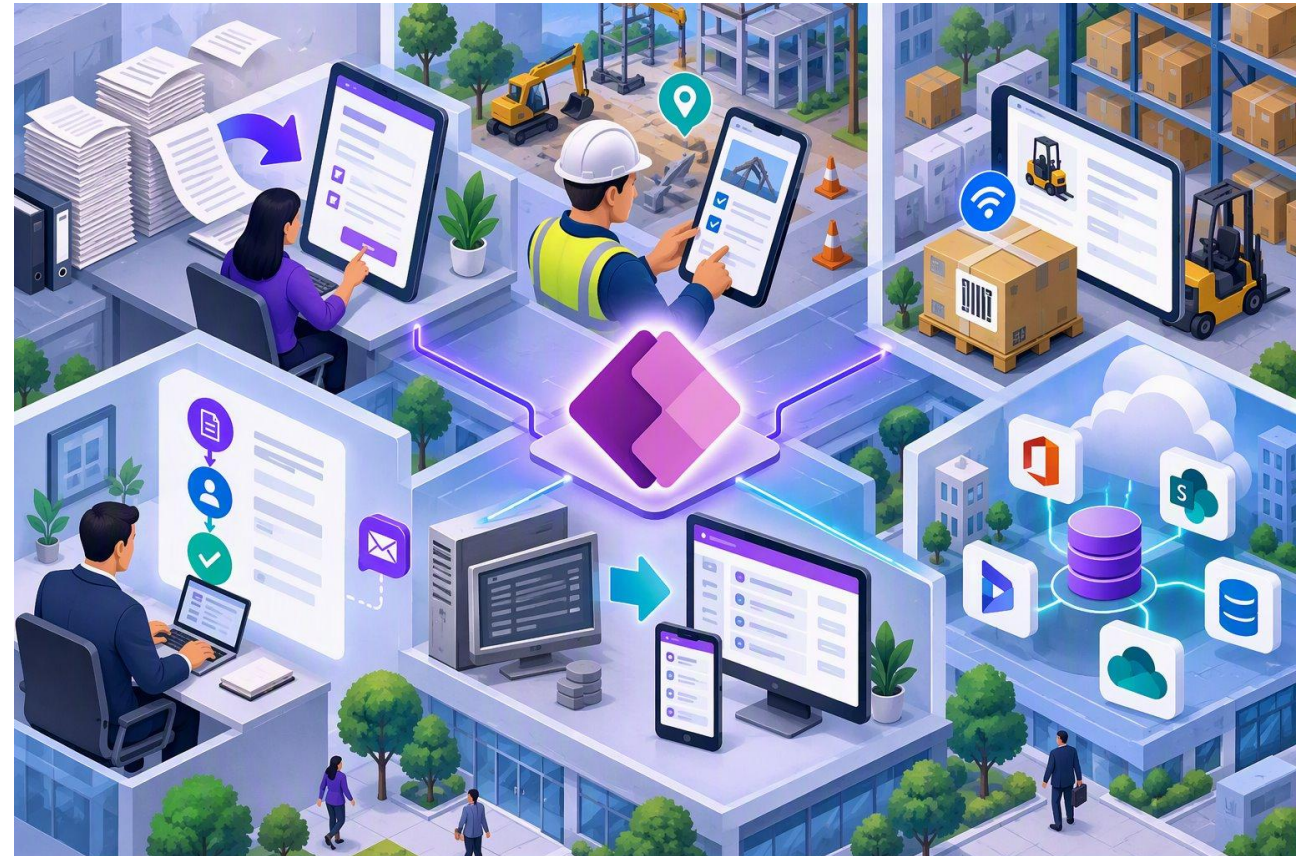
**BEST WHEN** Layout and interaction matter

**TYPICAL FIT** Mobile and task-focused apps

**START WITH** The data model

**BEST WHEN** Processes are data-dense

**TYPICAL FIT** Case and record management



They can also work together in one solution.

# Power Platform components summary

- Power Apps
  - Canvas app provides the employee experience
  - Model-driven app provides the operations experience
- Dataverse
  - Stores and models the business data
  - Other connected data systems can also be used
- Power Automate
  - Handles approvals, notifications, and workflow steps



# Lab

## What will we build today?

One business solution, built progressively from data to app to automation.

- Experience** Build a canvas app for everyday users
- Model** Create Dataverse tables and relationships
- Process** Create a model-driven app for operations
- Automate** Add business logic and an approval flow
- Deliver** Test, publish, and share the solution



# Lab details

- Create a machine ordering app with Power Apps
  - Online workshop from App in a Day series of courses
  - <https://learn.microsoft.com/en-us/training/paths/power-apps-online-workshop/>
- Exercises
  - Create a machine ordering Power Apps canvas app
    - Estimated time: 1 hr 24 min
  - Create a Dataverse table for the machine ordering app
    - Estimated time: 1 hr 33 min
  - Create a model-driven Power Apps app for machine ordering
    - Estimated time: 1 hr 20 min
  - Integrate a Power Apps app with Power Automate
    - Estimated time: 1 hr 17 min

# Lab domain – Coffee machine ordering

- Employees request new coffee machine for break room every few years
- Organization constrained by lack of development resources
  - No traditional developers available to create custom solution
- Goal: build a customized app for machine ordering
  - Runs on the web and mobile devices
    - Streamline machine order and approval
- Solution: Use Power Platform to empower tech savvy business users to create app
  - Power Apps
    - Application UI and interaction logic
  - Dataverse
    - Stores coffee machine order data
  - Power Automate
    - Stores machine order data